

CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

Internship Project Report

Submitted By

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Domain

Data Analytics

Abstract

Credit card fraud has become a major challenge in the digital payment industry. The objective of this project is to detect fraudulent credit card transactions using Machine Learning techniques. The dataset contains transaction details classified as either legitimate or fraudulent. Data preprocessing, exploratory data analysis (EDA), feature engineering, and classification modeling were performed. Logistic Regression was used as the classification algorithm with balanced class weights to handle class imbalance. The model successfully identifies suspicious transactions and helps reduce financial losses caused by fraudulent activities.

1. Introduction

With the rapid growth of online transactions and digital payments, credit card fraud has become a significant concern for financial institutions. Detecting fraudulent transactions manually is difficult because of the large volume of daily transactions. Machine Learning techniques provide an effective solution by automatically identifying unusual transaction patterns.

This project develops a fraud detection system that classifies transactions as legitimate or fraudulent based on transaction features.

2. Objectives

- To analyze credit card transaction data.
- To identify patterns associated with fraudulent transactions.
- To perform data preprocessing and feature engineering.
- To build a machine learning model for fraud detection.
- To evaluate model performance using classification metrics.
- To reduce financial losses caused by fraudulent activities.

3. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib

4. Dataset Description

The project uses a Credit Card Fraud Detection dataset containing transaction information.

Important Features

- Time
- Amount
- V1 to V28 (anonymized transaction features)
- Class

Target Variable

- Class = 0 → Legitimate Transaction
- Class = 1 → Fraudulent Transaction

5. Methodology

Step 1: Data Collection

The dataset was imported using Pandas and loaded into the Jupyter Notebook environment.

Step 2: Data Exploration

The following analyses were performed:

- Dataset shape inspection
- Feature information analysis
- Missing value detection
- Fraud vs legitimate transaction count
- Statistical summary of transaction amounts

Step 3: Exploratory Data Analysis (EDA)

Several visualizations were created:

- Transaction Class Distribution
- Fraud vs Legitimate Percentage Distribution
- Transaction Amount Histogram
- Box Plot of Amount vs Fraud Status
- Fraud Transactions Over Time
- Correlation Heatmap

Step 4: Feature Engineering

A new feature named:

$\text{Amount_log} = \log(\text{Amount} + 1)$

was created to normalize the transaction amount distribution.

Step 5: Data Preprocessing

- Features and target variables were separated.
- Train-test split was performed.
- StandardScaler was used for feature scaling.
- ColumnTransformer was applied for preprocessing.

Step 6: Model Building

A Logistic Regression model was implemented using Scikit-Learn Pipeline.

Model Parameters:

- Logistic Regression
- Class Weight = Balanced
- Max Iterations = 1000

Step 7: Model Evaluation

The model was evaluated using:

- Classification Report
 - Confusion Matrix
 - Accuracy Score
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6. Implementation

Libraries Imported

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib

Machine Learning Pipeline

1. Data Loading
 2. Data Cleaning
 3. Exploratory Data Analysis
 4. Feature Engineering
 5. Train-Test Split
 6. Data Scaling
 7. Logistic Regression Training
 8. Prediction
 9. Evaluation
 10. Model Saving
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7. Results

The machine learning model successfully classified transactions into legitimate and fraudulent categories.

Key outcomes:

- Fraudulent transaction patterns were identified.
 - Class imbalance was handled using balanced class weights.
 - The Logistic Regression model achieved high prediction performance.
 - Confusion Matrix and Classification Report were used to assess model effectiveness.
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8. Conclusion

This project demonstrates the use of Machine Learning in detecting credit card fraud. Through data preprocessing, exploratory analysis, and Logistic Regression modeling, fraudulent transactions can be identified effectively. Such systems help banks and financial institutions reduce losses and improve transaction security.

9. Future Scope

- Implement advanced algorithms such as Random Forest and XGBoost.
 - Use Deep Learning techniques for improved accuracy.
 - Develop a real-time fraud detection system.
 - Deploy the model using Flask or Streamlit.
 - Integrate with banking applications for live transaction monitoring.
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10. References

1. Scikit-Learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Credit Card Fraud Detection Dataset
7. Machine Learning Research Articles on Fraud Detection